

Flowchart

Task 1:

Draw a flowchart that reads two integers from the user and prints “first” if the first number is greater than the second number.

Task 2:

Draw a flowchart that reads two integers from the user. Your program should then print “first is greater” if the first number is greater, and “first is not greater” otherwise.

Task 3: [Homework]

Draw a flowchart that reads two integers from the user. Your program should then print “first is greater” if the first number is greater, “second is greater” if the second number is greater, and “the numbers are equal” otherwise.

Task 4: [Homework]

Draw a flowchart that reads two integers from the user, subtracts the smaller number from the larger one, and prints the result.

Task 5:

Draw a flowchart that reads an integer from the user, and prints “The number is even” or “The number is odd”, depending on whether the number is even or odd. (Hint: use the modulus operator)

Task 6: [Homework]

Draw a flowchart that reads an integer from the user, and if the number is even and greater than 10, prints “An even number greater than 10”. If the number is even but lesser than 10, print “An even number not greater than 10”. If the number is greater than 10 but odd, print “An odd number greater than 10”. If the number is odd and also less than 10, print “An odd number less than 10”.

Task 7: [Homework]

Draw a flowchart that reads an integer from the user, and prints the integer if it is a multiple of **either 2 or 5**. For example, 2, 4, 5, 6, 8, 10, 12, 14, 15, 16, 18, 20, 22 ...

Task 8:

Draw a flowchart that will take a number from the user and, if the number is divisible by 5, find 25% of the number. Otherwise, find 40% of the number. Then, add 20 to the resulting value and print the final result.

Task 9:

Write a java program that reads a student's mark for a single subject, and prints out the corresponding grade for that mark. The mark ranges and corresponding grades are shown in the table below.

Marks	Grade
97-100	A+
90-96	A
85-89	A-
70 - 84	B
57 - 69	C
50 - 56	D
<50	F

Variables and Data type

Task 1: [Homework]

What will be the output of the following lines? Try to determine the output **without writing any code**.

- a) "hello" + "goodbye"
- b) "3" + "5"
- c) "3 + 5 is " + 3 + 5
- d) "3 + 5 is " + (3 + 5)
- e) 3 + 5 + "is 3 + 5"
- f) (3 + 5) + "is same as 3 + 5"

To **verify your outputs**, create a class and a main method. Inside the main method, print the above statements and observe the outputs.

Intro to Java

Task 1: You are given a 5-digit integer number. Write a Java program to extract the last 2 digits and the first two digits of the given number. Now consider the first two digits of the number as the radius of a circle. Finally, calculate and print the circle's area and circumference.

Given Value: 34512	Sample Output: First two digits: 34 Last two digits: 12 Area: 3631.6811075498013 Circumference: 213.62830044410595
Given Value: 76165	Sample Output: First two digits: 76 Last two digits: 65 Area: 18145.839167134644 Circumference: 477.5220833456485

Operators

Task 1:

```
double d = 21;
int x = 5;
int c = 2;
x += 8;
c = x % c++;
System.out.println(c * 2);
d = 1 + d / c + 21;
System.out.println(d++ / 2 + 3);

int a = 12, b = 7, c = 20, d = 5;
System.out.println((((a + b * 2 - c / 4) % 3 > 2 && (d + a - b) < c) || ((a % b == 5) && (++c / d > 3))));
```

```
int a = 10, b = 5, c = 20;
System.out.println("Result: " + ((a + b - c) % 3 / 2 > 1 && a != b || c < 30 == true));

System.out.println((true && false) || (!false && (true || false)) && (false || true));

System.out.println((true || false && !false) && !(false || true) || (true && !false) || (false && true)) ||
(!true && false) );
```

Branching

1. Write a java program that reads a student's mark for a single subject, and prints out the corresponding grade for that mark. The mark ranges and corresponding grades are shown in the table below. **Valid Marks are 0 - 100** [Homework]

Marks	Grade
97-100	A+
90-96	A
85-89	A-
70 - 84	B
57 - 69	C
50 - 56	D
<50	F

2. Write a code that takes an hour from the user as input and tells it is time for which meal. Users will input the number in 24-hour format, say 14 to mean 2pm, 3 to mean 3am, 18 to mean 6pm etc. **Valid inputs are 0 to 23**. Inputs less than 0 or more than 23 are invalid in 24-hour clock.

Input will be whole numbers. For example, 3.5 will NOT be given as input.

Inputs: Message to be printed

4 to 6: Breakfast

12 to 13: Lunch

16 to 17: Snacks

19 to 20: Dinner

For all other valid inputs, say "Patience is a virtue"

For all other invalid inputs, say "Wrong time"

For example,

If the user enters 4, your program should print the message "Breakfast".

If the user enters 5, your program should print the message "Breakfast".

If the user enters 6, your program should print the message "Breakfast".

If the user enters 0, your program should print the message "Patience is a virtue".

If the user enters 1, your program should print the message "Patience is a virtue".

If the user enters 18, your program should print the message "Patience is a virtue".

If the user enters 23, your program should print the message "Patience is a virtue".

If the user enters 24, your program should print the message "Wrong Time".

If the user enters -1, your program should print the message "Wrong Time".

If the user enters 27, your program should print the message "Wrong time".

3. Convert all the flowchart problems (1-9) into Java code. **[Homework]**

4. Suppose you are hired by a supershop named Bastob. Now, your task is to create a Java program that will help the cashier calculate the change to be returned. The program takes two inputs. The first input is an integer number which is the amount of money to be paid in taka and the second integer is the amount of money the customer gave to the cashier.

Your program should print the following:

- If the customer gave more money than the actual amount, print change the cashier should return in notes and coins.
- If the customer gave less money than the amount to be paid, then print the amount the customer needs to pay.

Consider the following denomination for notes and coins in taka:

Notes: 100, 50, 20, 10.

Coins: 5, 2, 1.

Sample Input	Output
Enter the amount the customer need to pay(Taka) 35 Enter the amount, customer gave(Taka) 53	The returned amount is 18 taka. 100 taka note: 0 50 taka note: 0 20 taka note: 0 10 taka note: 1 5 taka coin: 1 2 taka coin: 1 1 taka coin: 1
Enter the amount the customer need to pay(Taka) 60 Enter the amount, customer gave(Taka) 500	The returned amount is 440 taka. 100 taka note: 4 50 taka note: 0 20 taka note: 2 10 taka note: 0 5 taka coin: 0 2 taka coin: 0 1 taka coin: 0
Enter the amount the customer need to pay(Taka) 50 Enter the amount, customer gave(Taka) 50	The returned amount is 0 taka.
Enter the amount the customer need to pay(Taka) 550 Enter the amount, customer gave(Taka) 520	Please pay 30 taka more.

5. Draw a flowchart and Write a java program that reads an integer, and prints the integer if it is **NOT** a multiple of 2 **OR** **NOT** a multiple of 5. **[Homework]**
6. Write a Java code of a program which asks user's age as an input. If his/her age is below 16, print "Too young to drive"; if his/her age is exactly 16, print "Better clear the road" and if his/her age is above 16, print "You are getting pretty old". **[Homework]**

7. What will be the output of the following program? Your answer will not be accepted without the workings.

1	<code>public class Tracing3 {</code>
2	<code> public static void main(String[] args) {</code>
3	<code> int p = 5;</code>
4	<code> int q = 6;</code>
5	<code> int r = 9;</code>
6	<code> int sum = 0;</code>
7	<code> if (p < 12) {</code>
8	<code> System.out.println(r + 2);</code>
9	<code> }</code>
10	<code> else {</code>
11	<code> System.out.println(r + p);</code>
12	<code> }</code>
13	<code> if (q > 20){</code>
14	<code> System.out.println(r + 19);</code>
15	<code> }</code>
16	<code> else if (q <= 6) {</code>
17	<code> System.out.println(q + 3);</code>
18	<code> }</code>
19	<code> else{</code>
20	<code> System.out.println(p + q + r);</code>
21	<code> }</code>
22	<code> if (r > 15) {</code>
23	<code> System.out.println(r);</code>
24	<code> }</code>
25	<code> else if (r == 0) {</code>
26	<code> System.out.println(p + q);</code>
27	<code> }</code>
28	<code> else {</code>
29	<code> System.out.println(p);</code>
30	<code> }</code>
31	<code> if (sum != 0) {</code>
32	<code> System.out.println(3);</code>
33	<code> }</code>
34	<code> else {</code>
35	<code> System.out.println(sum + 32);</code>
36	<code> }</code>
37	<code> if(p > 0 && r < 10){</code>
38	<code> System.out.println(p + r);</code>

39	}
40	else {
41	System.out.println(p - r);
42	}
43	}
44	}

Loops

1. Print (1-100) using both for and while loop
2. Print only the numbers which are divisible by 4 in between (20-100).
3. Print the following sequences:
 - a) 2 4 6 8 10
 - b) 0 3 6 9 12
4. Take 5 numbers from the user and print the average of the numbers.
5. Take 10 numbers from the user and count how many numbers are even and odd
6. Take 15 numbers from the user and find the maximum & minimum number.
7. Take an integer number from the user and count & print the total number of digits of that number.
8. Take an integer number from the user and print the digits of that number.
9. Take a number from the user and count and print the divisors of that number.
10. Take a number from the user and check whether the number is a prime number or not. If prime, print "Prime". Otherwise print "Not Prime".
11. Take a number from the user and check whether the number is a prime number or not. If prime, print "Perfect". Otherwise print "Not Perfect".
12. Write a java program that keeps on taking user inputs until the user provides 0 and prints the maximum, minimum, and average of all the even positive numbers given by the user. If no even positive number is given, the average should be zero.

Sample Input	Sample Output
12 -8 19 8 -1	Max: 12 Min: 8 Average: 10

0	
Explanation: After giving 5 numbers, the user provided 0, therefore, no more inputs were taken. Among these, only 12 and 8 are even positive numbers.	

13. Write a Java program to take a positive integer N (where $N > 0$) as user input and print the **first N prime numbers starting from 2**. Your code should check all the positive integers starting from 2 and determine whether they are prime or not until N prime numbers are found.

Sample Input 1:

5

Sample Output 1:

2

3

5

7

11

Sample Input 2:

7

Sample Output 2:

2

3

5

7

11

13

17

14. [Homework]

1	<code>public class Trace2 {</code>
2	<code> public static void main(String[] args) {</code>
3	<code> int x = 0, p = 0, sum = 0;</code>
4	<code> p = 1;</code>
5	<code> x = 2;</code>
6	<code> double q = 0.0;</code>
7	<code> sum = 5;</code>
8	<code> while (p < 15) {</code>
9	<code> q = x + p - (sum + (int) (7 / 4)) / 3.0 % 2;</code>
10	<code> sum = sum + x + (int) q;</code>
11	<code> x += 1;</code>
12	<code> System.out.println(sum);</code>
13	<code> if (x > 5) {</code>
14	<code> p += (int) (5 / 2);</code>
15	<code> }</code>
16	<code> else {</code>
17	<code> p += 10 % 3;</code>
18	<code> }</code>
19	<code> }</code>
20	<code> sum = sum + p;</code>
21	<code> System.out.println(sum);</code>
22	<code> }</code>
23	<code>}</code>

15. [Homework]

1	<code>public class T1{</code>
2	<code> public static void main(String args[]){</code>
3	<code> int x = 3, y = 0;</code>
4	<code> int sum = 0;</code>
5	<code> while (x < 6){</code>
6	<code> y = x - 3;</code>
7	<code> while (y < 3){</code>
8	<code> sum = (sum % 3) + x - y * 3 ;</code>
9	<code> System.out.println(sum);</code>
10	<code> y = y + 1;</code>
11	<code> }</code>
12	<code> if (x > 5){</code>
13	<code> x++;</code>
14	<code> }</code>
15	<code> else{</code>
16	<code> x += 2;</code>
17	<code> }</code>
18	<code> }</code>
19	<code> }</code>
20	<code>}</code>

16. [Homework]

1	public class Task21{
2	public static void main(String[] args) {
3	int m = 17, n = 13, p = 1, sum = 30;
4	while(0<p%10){
5	if(m % 10 == 0){
6	sum = sum * m % n + p / n ;
7	}
8	else{
9	if(m % 4 == 0){
10	sum += n + (--m);
11	}
12	else{
13	sum -= m--;
14	}
15	}
16	p+=1;
17	System.out.println(sum++);
18	}
19	System.out.println(!(n%13 == 0) && !false p>10);
20	}
21	}

String

1. [Classwork]

Write a Java program that takes a String as input from the user and prints that String in reverse order. Also check if it's a palindrome or not.

Sample Input 1:

CSE110

Sample Output 1:

011ESC

Not a palindrome

Sample Input 2:

madam

Sample Output 2:

madam

Palindrome

Sample Input 3:

1234321

Sample Output 3:

1234321

Palindrome

2. [Homework]

Write a Java program that takes 2 inputs from the user, where the first input is a string with length greater than 1. The second input is the index of the first given string from where you have to start reversing. After reversing the first input string from that index, print the new string back to the user. See samples below for clarification.

Sample Input 1:

72418

4

Sample Output 1:

81427

Explanation: Our second input, index '4' is the last index of our first input String '72418', hence the entire string is reversed giving us '81427'.

Sample Input 2:

12345

2

Sample Output 2:

32145

Explanation: The second input is '2' so we have to reverse from index 2 of our first input. The 2nd index of our first input String is '3', index 1 is '2' and index 0 is '1'. Hence, if we reverse indexes 0 to 2, we get '321'. Index 3 and 4 which are '4' and '5' respectively remain unchanged hence our final output is '32145'

3. [Classwork]

Write a program that will take a string from the user and count how many uppercase, lowercase, digits, and special characters are in the string.

Sample Input:

My name IS bob@23!

Sample Output

Uppercase = 3

Lowercase = 8

Digit = 2

Special characters = 4

4. [Classwork]

Write a program that will take a string and a character from the user and check if the character is in the string or not. If the character is in the string then print the index number of the character there it is found first. Otherwise, print 'Character not found'.

[To take a character input from the user use: `sc.next().charAt(0)`]

Sample Input:

Bangladesh

a

Sample Output

1

Sample Input:

Java Language

p

Sample Output

Character not found

5. [Homework]

Write a program that will take a string from the user and count how many special characters are in the string. Consider only these `#!?&` special characters. That means if the input string contains other special characters than `@#!?&`, then your program won't count them.

Sample Input:

<jAVa is# 2fun & I love Java!+

Sample Output

Special characters = 3

6. [Homework]

Write a java program that will ask a string from the user and count how many vowels and consonants are there in the string.

Sample Input:

I love Bangladesh

Sample Output

Vowel = 6

Consonants = 9

7. [Classwork]

Write a program that will take an equation of simple operation of two integers as a string as input from the user and perform that operation.

Sample Input 1:

234+678

Sample Output 1:

912

Sample Input 2:

87531*123

Sample Output 2:

10766313

8. [Homework]

Write a Java program that takes a person's first name and email's password as user inputs and checks whether that password is valid or not. A password will be considered invalid if the person's name is present anywhere in that password. *You can assume that both the name & password will not contain any uppercase characters and the length of the password will always be greater than the length of name.*

[Note: You are not allowed to use the built-in contains() or substring() method to solve this problem.]

Sample Input	Output
dipu iamdipu123#	Invalid
shajib happy new year 2025	Valid

9. [Homework]

Write a Java program that takes a sentence as user input and displays the sentence in aLtErNaTiNg CaPs format. Note that, you have to ignore spaces and other punctuations while altering the characters. Also, your new sentence will always start with lowercase letters.

Sample Input	Output
Hello World!	hElLo WoRlD!

I love Java String Problems. Those are easy.

i LoVe JaVa StRiNg PrObLeMs. ThOsE
aRe EaSy.

Array

```
public class A{
    public static void main(String args[]){
        int [] myArray = {3, 9, 33, 9, -2};
        double [] b = {5.5,3.5,-4.0};
        int index1 = 0, index2 =0;
        index1 = 1;
        b = myArray;
        while (index1 < 5){
            myArray[index1] = index1 + 1;
            index2 = 1;
            while (index2 < index1 ){
                myArray[index1] = b[index2 - 1] + myArray[index2] - index1;
                index2 = index2 + 1;
            }
            System.out.println(myArray[index2]);
            index1 = index1 + 1;
        }
    }
}
```

Method

- A. Write a method called **evenChecker** that takes an **integer** number as its argument and prints whether the number is even or odd **inside the method**.

Sample Method Call	Sample Output
evenChecker(10);	Even!!
evenChecker(17);	Odd!!

- B.** Write a method called **isEven** that takes an **integer** number as an argument and **returns** boolean true if the number is even otherwise **returns** boolean false.

Sample Method Call	Sample Output
<pre>boolean result = isEven(10); System.out.println(result);</pre>	true
<pre>boolean result = isEven(17); System.out.println(result);</pre>	false

Recursive Method

Trace the following code to generate the outputs. Show the necessary trace table.

1	public class Trace2{
2	public static void main(String [] args){
3	methodA(5) ;
4	}
5	public static void methodA(int n){
6	if (n>=1){
7	methodA(n-1) ;
8	System.out.println(methodB(n)) ;
9	}
10	}
11	public static int methodB(int a){
12	int b = (++a) + 6;
13	return b*a-a;
14	}
15	}

Trace the following code to generate the outputs. Show the necessary trace table.

1	public class Trace3 {
2	public static void main(String[] args) {
3	int[] arr = {22, 7, 14, 33, 19};
4	int result = myMethod(arr, 0);
5	System.out.println(result);
6	}
7	public static int myMethod(int[] a, int idx) {
8	if (idx == a.length - 1) {
9	return a[idx];
10	}
11	int num = myMethod(a, idx + 1);
12	System.out.println(a[idx] + num);
13	if(a[idx] < num) {
14	return a[idx];
15	}
16	else{
17	return num;
18	}
19	}
20	}

Trace the following code to generate the outputs. Show the necessary trace table.

1	public class Tracel{
2	public static void main(String [] args){
3	int[] arr = {7, 2, 5};
4	for (int idx = 0; idx < arr.length; idx++){
5	System.out.println(findFibonacci(arr[idx]));
6	}
7	}
8	public static int findFibonacci(int n){
9	if (n < 2){
10	return n;
11	}
12	else{
13	return findFibonacci(n - 2) + findFibonacci(n - 1);
14	}
15	}
16	}

Basic Loop

Write a Java program that will take a number from the user and check if the digits are in **decreasing order from right to left**. If all the digits are in decreasing order, print “Decreasing order”. Otherwise, print “Not in decreasing order” along with the first number for which the order is not maintained.

User Input	Sample Output
13469	Decreasing Order
6734	Not in decreasing order 3 is not greater than 7
654312	Not in decreasing order 1 is not greater than 3

Array + String

You are given a String array that contains **positive** temperature values in **Celsius** or **Fahrenheit**, represented as Strings. Write a Java program that creates and prints two

new **integer** arrays: one containing only the Fahrenheit temperatures and the other containing only the Celsius temperatures from the given array. You can't use any built-in methods other than `.length()`, `.charAt()`, `.toString()`. **[Your code should work for any given String array of any length]** **[Mark: 10]**

Given Array	Sample Output
String[] arr = {"32F", "0C", "212F", "100C", "50F"};	Fahrenheit: [32, 212, 50] Celcius: [0, 100]
String[] arr = {"14F", "10F", "232C", "67F", "50C"};	Fahrenheit: [14, 10, 67] Celcius: [232]

You are given a **String** array which contains some positive and negative numbers in String format. Write a Java program that will create and print two new **integer** arrays: one containing only the positive numbers and the other containing only the negative numbers from the given array. You can't use any built-in methods other than `.length()`, `.charAt()`, `.toString()`. **[Your code should work for any given String array of any length]**

Given Array	Sample Output
String [] arr = {"+23", "-14", "-122", "+20", "-100"};	Positive: [23, 20] Negative: [-14, -122, -100]
String [] arr = {"+223", "-194", "+0", "+4420"};	Positive: [223, 0, 4420] Negative: [-194]

String + Method

Write a Java method called **stringGenerator()** that takes a string as an argument and creates a new string. For each character in the string, add the character followed by the sum of the following:

- The ASCII value of the character.
- The number of times that character has appeared so far in the string (including the current occurrence).

Finally, print the new string during the method call. For simplicity, assume that the argument string will only contain lowercase letters.

Method Call	Sample Output
String ans= stringGenerator("abcdaa") System.out.println(ans)	a98b99c100d101a99a100
String ans= stringGenerator("aabca") System.out.println(ans)	a98a99b99c100a100

String + Method + Array Tracing

1	public class A{
2	public static void main(String [] args){
3	int arr [] = {1,2,3};
4	metB(arr);
5	double a = metA(5) + metC(12,23);
6	System.out.println(a);
7	System.out.println("End");
8	}
9	public static double metC(int a, double b){
10	b = a + b + metA(2);
11	System.out.println(b);
12	System.out.println("metA");
13	return b;
14	}
15	public static double metA(int a){
16	int ch = '1';
17	a+=14;
18	System.out.println(a+ch);
19	return a;
20	}
21	public static void metB(int arr []){

22	arr[0]+=arr[2];
23	System.out.println(arr[0]);
24	System.out.println("HelloWorld".charAt(arr[0]));
25	}
26	}

Method Tracing

1	public class A{
2	public static int metA(int n) {
3	if (n <= 5){
4	return 2;
5	}
6	else{
7	System.out.println(n-2);
8	return n + metB(n - 1);
9	}
10	}
11	public static int metB(int x) {
12	System.out.println(x);
13	if (x <= 5){
14	return 1;
15	}
16	else{
17	System.out.println(x + 4);
18	return x + metA(x - 1);
19	}
20	}
21	public static void main(String[] args) {
22	int result = metA(10);
23	}
24	}
25	}

Arrays + Strings + Methods

A. Write a method called **isVowel** which takes a string in its parameter and counts all the vowels in the String. If any vowel exists in the string then the method returns the **count**.

Sample Input	Sample Output
The quick brown fox jumps over the lazy dog	Number of vowels in the string: 11

B. Write a method called **isConsonant** which takes a string in its parameter and counts all the consonants in the String. If any consonant exists in the string then the method returns the **count**.

Sample Input	Sample Output
The quick brown fox jumps over the lazy dog	Number of consonants in the string: 24

C. Write a method called **vowel/consonantSum** which takes an array of strings in its parameter and returns the summation of the number of vowels/consonants.

Note: **You must call** the methods written in tasks **A/B**, otherwise this task will be **considered invalid**.

Given Array	Sample Output
<pre>String [] names = {"Bob", "Alice", "Max", "Marry", "Rosy"}; System.out.println("The total number of vowels in the array is:" + vowelSum(names)); System.out.println("The total number of consonants in the array is:" + consonantSum(names));</pre>	The total number of vowels in the array is: 7 The total number of consonants in the array is: 13